

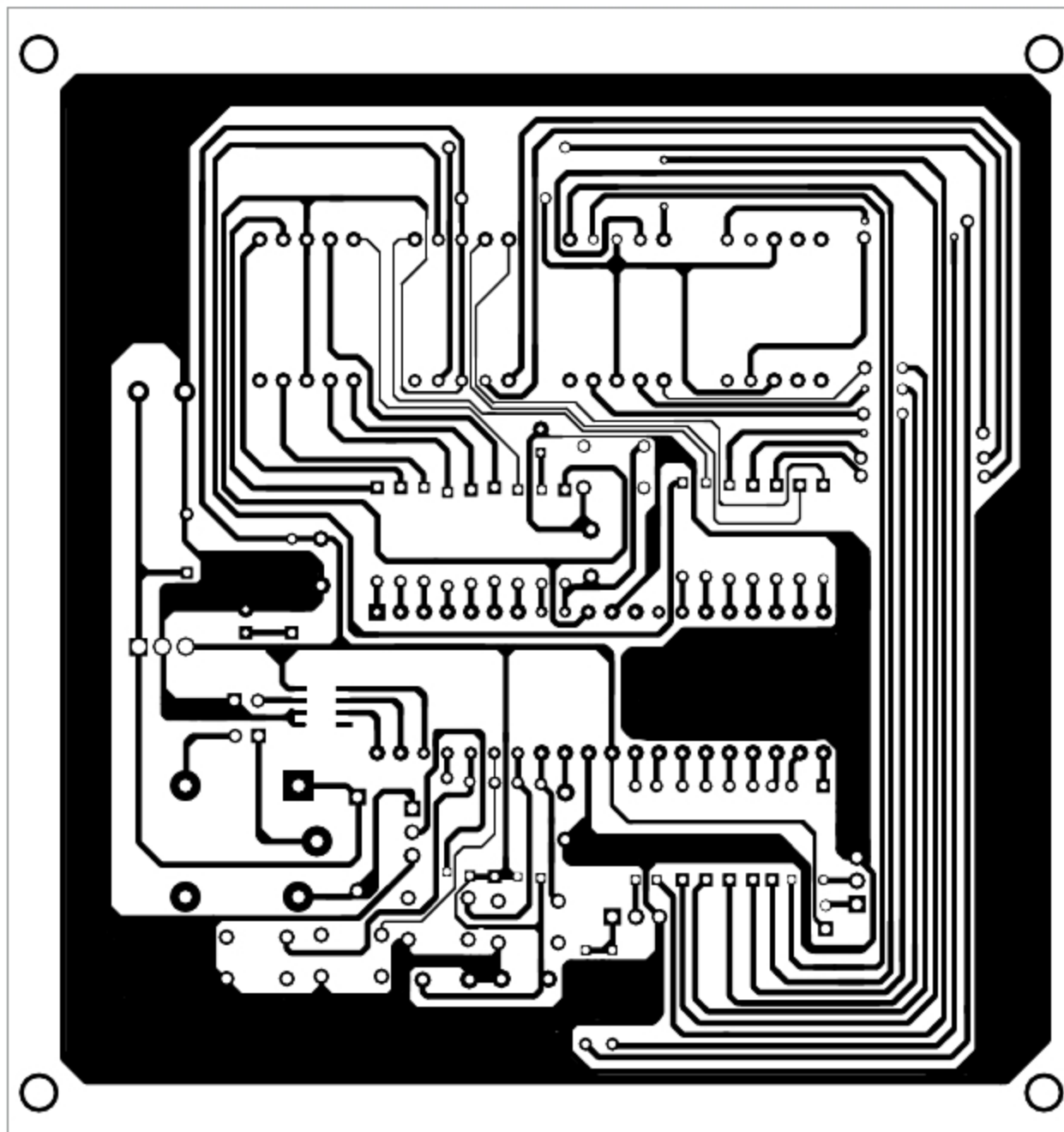


Fig. 2: Actual-size PCB layout of digital thermometer

other components.

The circuit generates an alarm through a buzzer, and a relay gets energised on crossing a pre-determined temperature value. Thus, this circuit regulates the heating device by switching it on or off through the relay. The heat source is connected between COM and NC pins of the relay.

You can also connect a draught fan between COM and NO pins of the relay. In normal operation, the relay will control the main relay (that is, high current capacity/hold-back relay), which controls the fuel or other inputs to the furnace/heater.

## Software

The firmware code (MAX6675thermo-meter.c) is written in C and compiled using AVR Studio. The hex code generated is burnt into ATmega16 using a suitable AVR programmer board.

Initially, when the circuit is switched on, the program makes all

four displays glow as a self-test. Then, the displays show an alarm temperature value for a while. Thereafter, the temperature is displayed continuously. While MAX6675 module is capturing temperature value, LED2 blinks continuously.

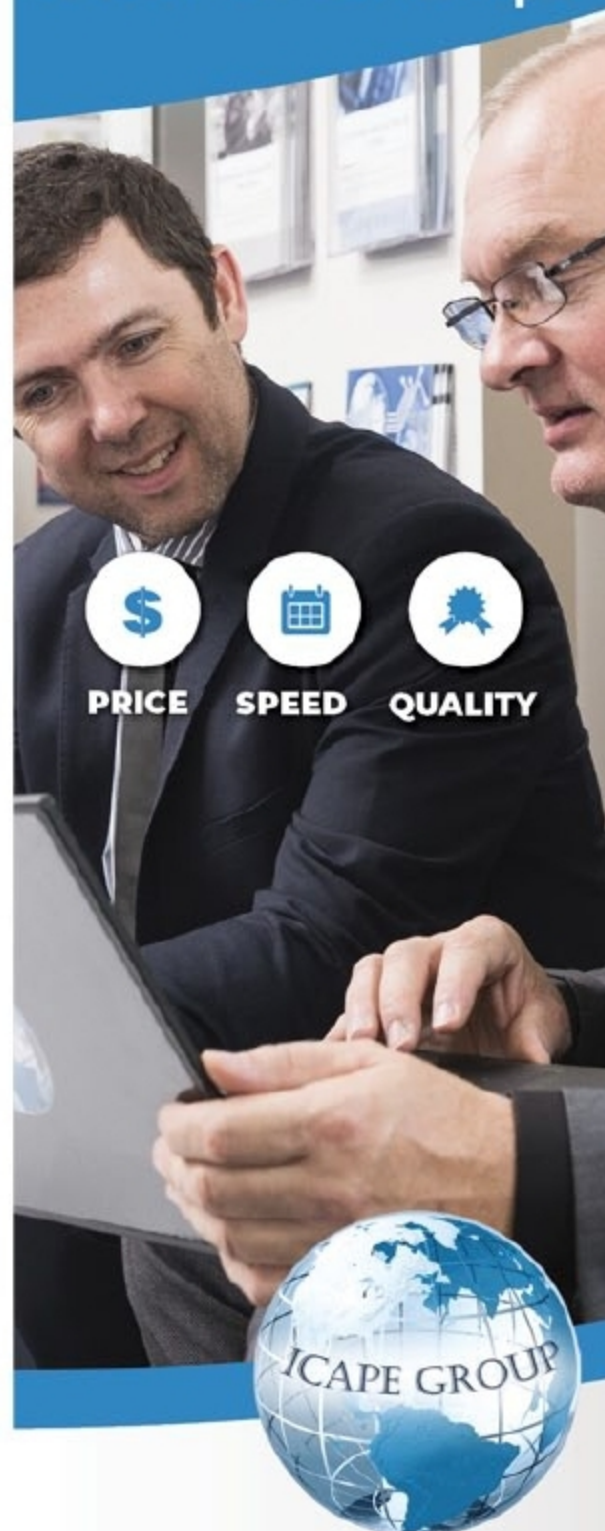
The circuit identifies open-loop condition through the displays. If thermocouple is not properly connected to MAX6675, OPE will be displayed. If there is an error with MAX6675 module, Err will be displayed.

## Construction and testing

An actual-size PCB layout of the industrial digital thermometer is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit on the PCB, connect a 12V battery across BATT.1.

To change the alarm or current temperature value, press switch S1 (alarm) or S2 (adj), respectively. If all four displays (DISP1 through DISP4)

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